

Project Objective

The objective of this project was to build an end-to-end customer churn prediction and risk segmentation system for a telecom business using machine learning techniques. The project aimed to identify churn-prone customers, analyze behavioral risk patterns, and generate actionable retention insights.

Dataset Overview

The analysis was performed on the Telco Customer Churn dataset containing customer demographic, billing, contract, and service-related information. After preprocessing and data quality corrections, the dataset was prepared for predictive modeling and customer risk analysis.

Key Findings

- Customers with month-to-month contracts showed significantly higher churn behavior.
- Short-tenure customers were more likely to discontinue services.
- Higher monthly charges were strongly associated with elevated churn probability.
- Customers without technical support exhibited increased churn risk.

Model Performance

Three machine learning models were evaluated:

- Logistic Regression
- Random Forest
- Gradient Boosting

Among them, Gradient Boosting achieved the strongest overall predictive performance based on ROC-AUC and Recall metrics.

Risk Segmentation Insights

Customers were segmented into:

- High Risk
- Medium Risk
- Low Risk

High-risk customers primarily consisted of users with short tenure periods, flexible contracts, and higher billing patterns.

Business Recommendations

1. Introduce targeted retention campaigns for high-risk customers using discounts and loyalty incentives.
2. Improve customer onboarding and technical support engagement during the early customer lifecycle stage.
3. Encourage long-term contract adoption to improve retention stability.

Conclusion

This project successfully demonstrated how predictive analytics and customer segmentation techniques can support telecom retention strategies and improve business decision-making.